Flood Hazard Risk Analysis for Urban Blocks Along Bogotá's Hydrology Network

Overview

This project evaluates flood hazard exposure for urban blocks (manzanas) in Bogotá, Colombia. The analysis combines two official hazard datasets, a hydrology network, and a 100-meter proximity buffer to classify each block into four levels of risk. The workflow was completed in two environments: a Jupyter Notebook using GeoPandas, and a custom ArcGIS Pro Python toolbox replicating the same geospatial logic. Both environments produced consistent spatial outputs and cartographic maps.

Data Description

All datasets used in this analysis come from official geospatial data providers in Colombia. These sources are widely recognized for producing authoritative environmental, hydrological, and urban planning datasets for Bogotá and the broader Cundinamarca region. The datasets were downloaded specifically for this exam and were not part of any prior course project.

Hazard Datasets (IDECA / IDIGER)

- Amenaza_IN_desbordamiento_20211229.shp
- Amenaza_IN_rompimiento_jarillon_20211229.shp

These two flood hazard layers were obtained from IDECA (Infraestructura de Datos Espaciales de Bogotá), the official spatial data portal managed by the Bogotá Mayor's Office, in coordination with IDIGER (Instituto Distrital de Gestión del Riesgo y Cambio Climático). The first represents flooding due to overflow ("desbordamiento"), while the second represents flooding from levee/dike failure ("rompimiento de jarillón").

Hydrology Dataset (IDECA)

- Drenaje.shp
  This polyline dataset was obtained from IDECA's hydrology and drainage infrastructure database. It includes Bogotá's rivers, creeks, irrigation channels, and stormwater pathways. It serves as the basis for creating the 100-meter hydrology buffer used to assess proximity-based flood risk.

Urban Blocks Dataset (IDECA / Secretaría Distrital de Planeación)

- MANZANA.shp
  Urban blocks ("manzanas") were obtained from the official cadastral and planning layers provided by IDECA and the Secretaría Distrital de Planeación (SDP). These polygons represent the basic administrative and urban design units used for land-use planning and socio-demographic analysis in Bogotá.

Boundary Dataset (IDECA)

- bogotapoly.shp
  The Bogotá boundary polygon was also sourced from IDECA. It defines the spatial extent of the Distrito Capital, including urban and rural areas. This dataset was used to clip all other layers to the appropriate study area.

Methods (Python Notebook)

The Jupyter Notebook begins by loading all datasets and merging the two hazard layers. All data are projected into EPSG:3116 (MAGNA-SIRGAS Bogotá), which ensures accurate buffering and overlay operations. The merged hazard layer, hydrology features, and blocks are then clipped to the Bogotá boundary to focus the analysis strictly on the urban zone.

A 100-meter buffer is created around hydrology lines, dissolved into a single polygon, and clipped to the study area. Spatial overlays are used to determine which hazard polygons intersect the hydrology buffer and to identify blocks intersecting hazard areas, buffer areas, both, or neither. A new attribute field **RISK_CAT** assigns each block into one of four categories based on this logic: Hazard & Buffer, Only Hazard, Only Buffer, None. Giving us four risk categories.

Two final maps are created using Matplotlib: (1) a context map showing all blocks with hazard polygons and hydrology, and (2) a risk-classification map showing blocks symbolized in four shades of orange.

Methods (ArcGIS Pro Python Toolbox)

A custom Python toolbox (*FloodingRisk.pyt*) was created to reproduce the full workflow inside ArcGIS Pro. The tool includes parameters for the two hazard inputs, the hydrology layer, blocks, the Bogotá boundary, the buffer distance, and the output feature class.

When the tool is executed, it performs the following steps: 1. Projects all inputs to EPSG:3116, 2. Merges the two hazard layers and assigns the HAZ_TYPE field, 3. Clips hazard polygons, hydrology lines, and blocks to the urban boundary, 4. Buffers the hydrology network by the user-defined distance, 5. Flags each block with IN_HAZARD and IN_BUFFER, 6. Computes the final RISK_CAT classification using the same four-category logic as the Notebook and 7. Outputs a feature class representing block-level flood exposure

The tool was validated, documented using ArcGIS Pro's Metadata Editor, and successfully packaged as a geoprocessing tool.

Results

The final outputs include both the Notebook-generated shapefiles and the ArcGIS Pro tool output. The two required maps clearly illustrate:

1. Flood hazard polygons, hydrology, and urban blocks (context map)
2. Block-level flood risk categories based on hazard intersection and hydrologic proximity

All map layouts include a legend, north arrow, and scalebar to communicate symbology and spatial scale.

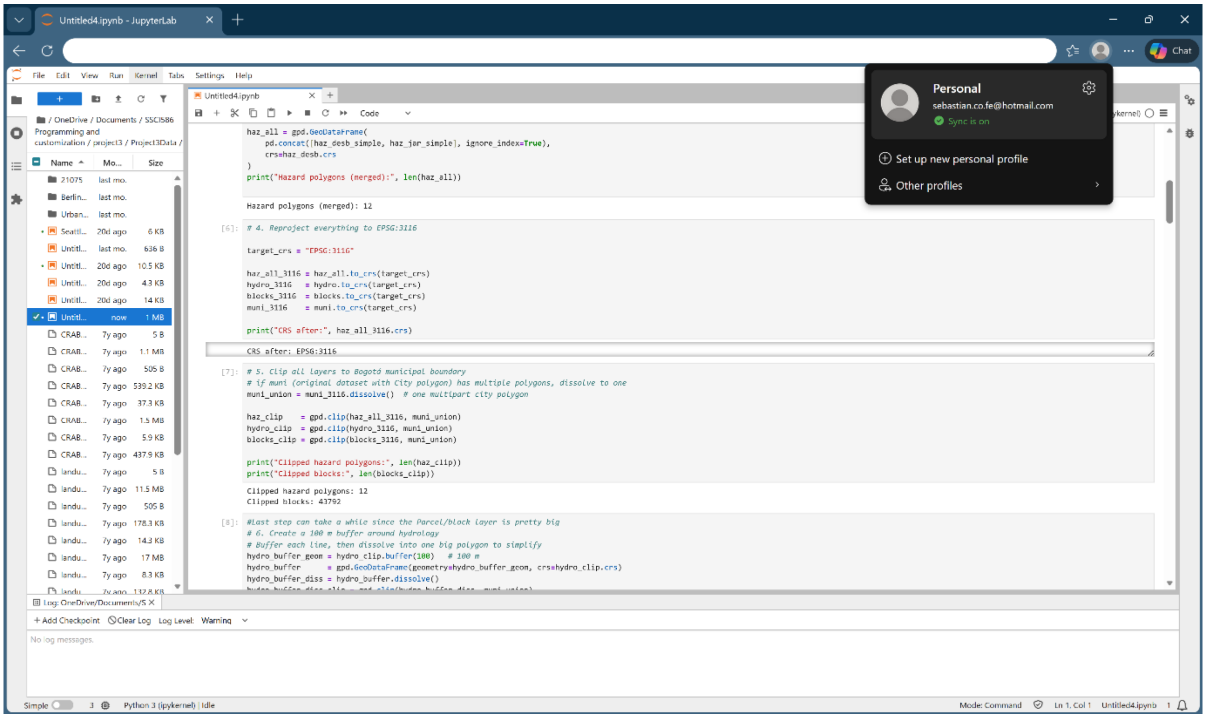
Summary

This project demonstrates the integration of geospatial data, Python scripting, and ArcGIS automation to perform a block-level flood hazard assessment for Bogotá. By reproducing the workflow in both Jupyter and ArcGIS Pro, the analysis shows proficiency in GIS programming, geoprocessing automation, cartographic design, and spatial risk assessment.

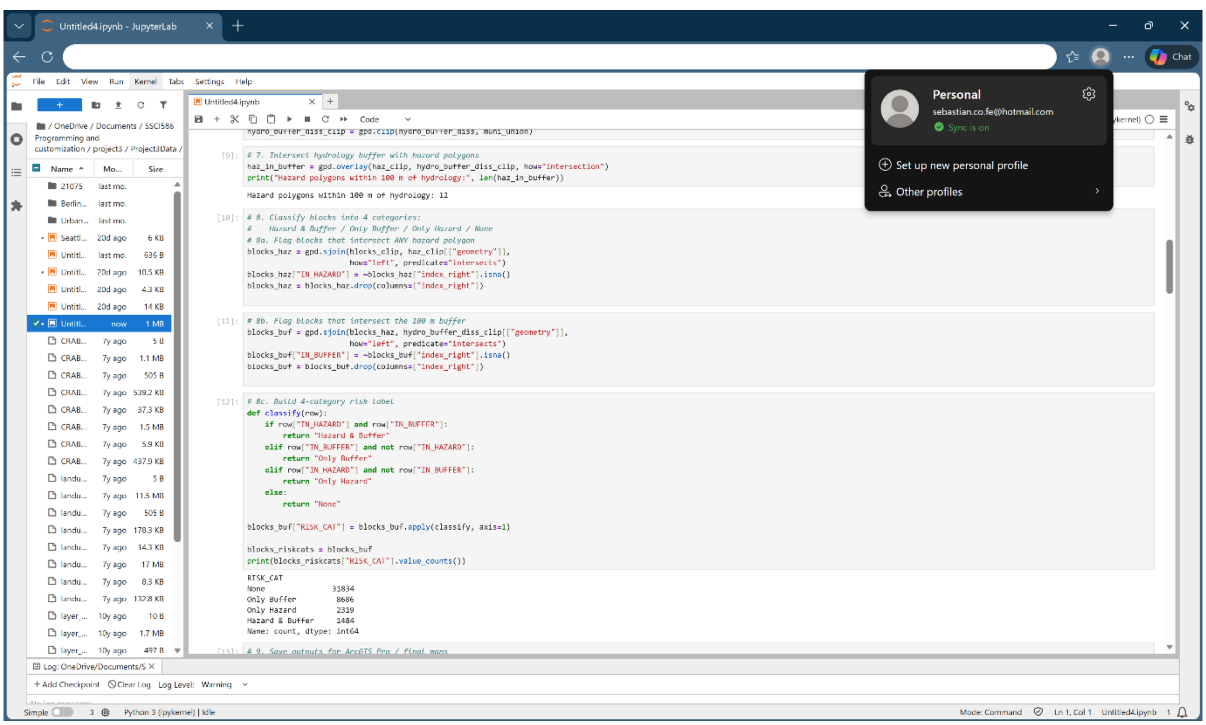
Screenshots Included

Four screenshots from the successfully executed Notebook:

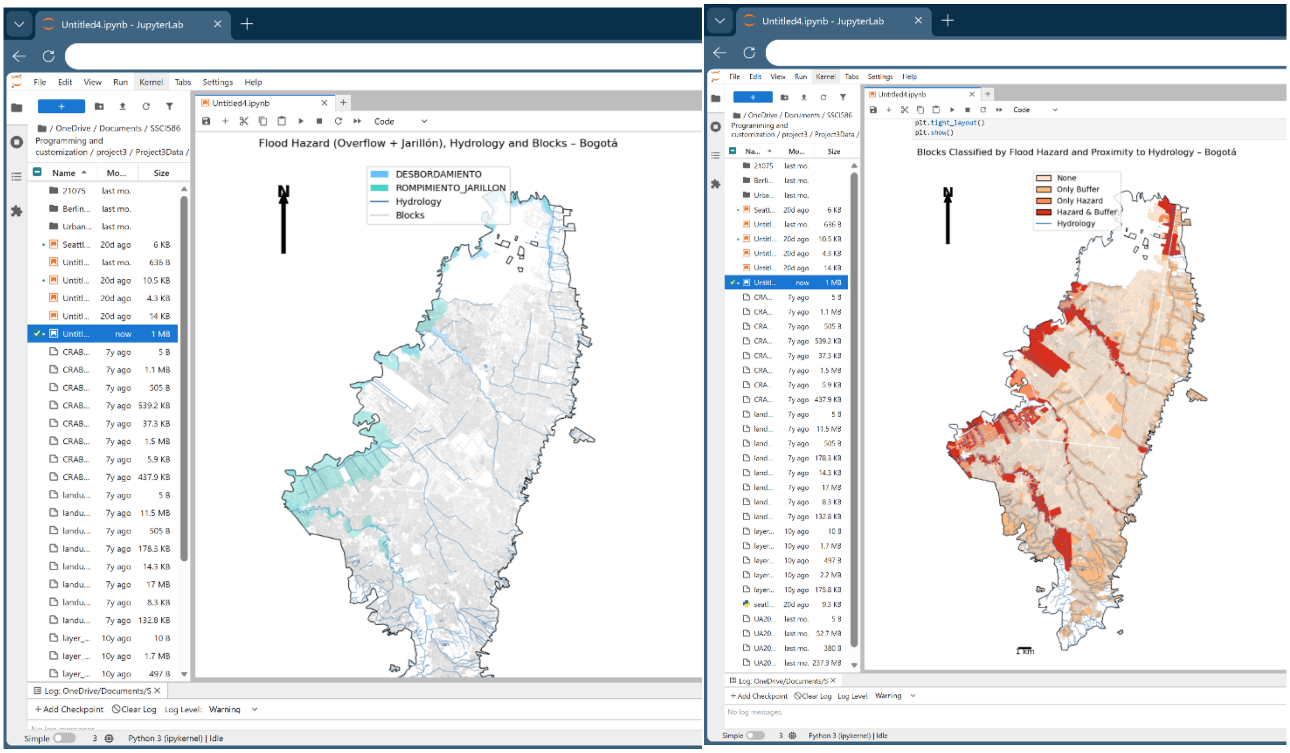
- Unified hazard creation



- Hydrology buffer generation and Block classification (RISK_CAT)

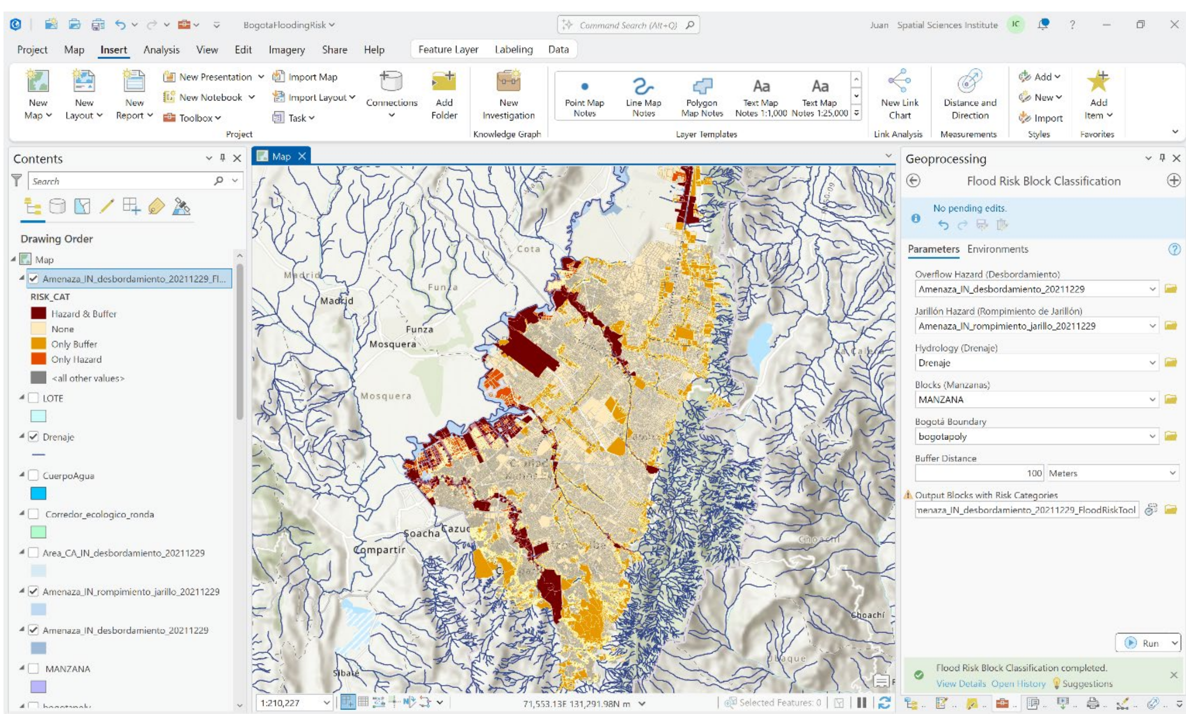


- Final Matplotlib maps



Two additional screenshots from ArcGIS Pro:

- The final layout map



- Successful tool execution in the Geoprocessing pane

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